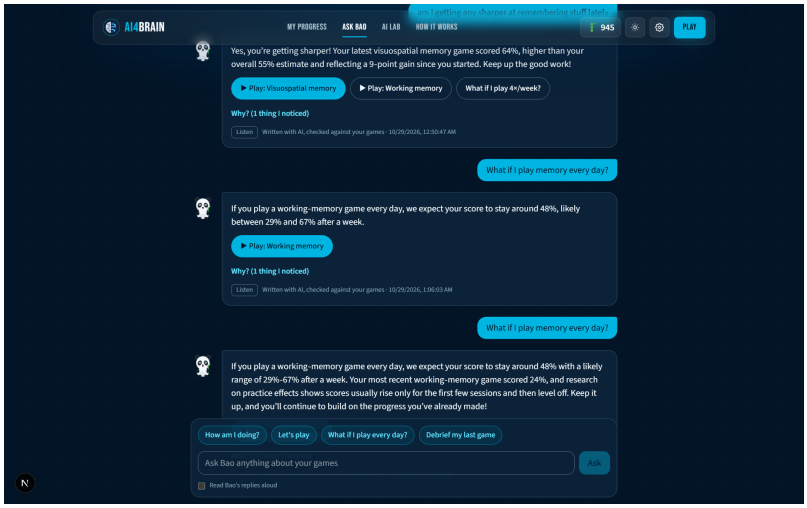
AI4Brain

An AI cognitive coach that can't invent a number

Anonymous submission

In one paragraph. AI4Brain combines three families of AI, each used only where it is the right tool. **Generative AI** (large language models) understands questions, writes the coach's explanations and powers two games that train people to judge persuasive and fabricating AI. **Probabilistic AI** (Bayesian inference and active learning) estimates seven cognitive skills with honest uncertainty and picks the most informative next game. **Guardrails** (deterministic verification and safety filters) sit between the models and the user, so a language model can never put a number, a diagnosis or an invented skill on screen.

7	2	89–95%	544
AI components, each with a fallback	model calls at most per coach answer	of 90% bands hold the true ability	automated tests, all passing



Bao, the AI coach, answering "What if I play memory every day?" in the running application.

Contents

1 Introduction: where AI belongs **2**

2 The AI map **2**

3 Why these AI choices **3**

4 The AI coach **3**

4.1 Step 1: the intent router . . . 3

4.2 Step 2: retrieval from the user's own data . . . 3

4.3 Step 3: the writer . . . 4

4.4 Step 4: the verifier . . . 4

4.5 When there is no AI . . . 5

5 AI that trains people to judge AI **5**

5.1 Second opinion: a simulated advisor . . . 5

5.2 The AI Lab: two games against a live model . . . 6

5.3 The AI Collaboration Score . . . 7

6 Perception AI: voice **7**

7 What each AI sees **7**

8 The measurement engine (probabilistic AI) **7**

8.1 From trials to an observation . . . 7

8.2 The Brain Twin . . . 8

8.3 Active learning and forecasts . . . 8

9 Validation **8**

10 Ethics and safety **9**

11 Limitations and next steps **10**

A Formulas **10**

1 Introduction: where AI belongs

A cognitive coach has two very different jobs. It must *measure*: turn a minute of play into a skill estimate that is accurate and honest about its uncertainty. And it must *talk*: understand what a person asks, explain results in plain words and keep them motivated. Today's language models are excellent at the second job and unreliable at the first: they write fluently, but they miscount, drift between runs and cannot attach a calibrated probability to an estimate.

AI4Brain is built on that observation. **Each kind of AI does the job it is good at, and code checks the seams between them**:

- **Generative AI** reads and writes language: it understands the user's message, drafts the coach's answer, and argues or fabricates on purpose in two training games.
- **Probabilistic AI** does the measuring: a Bayesian model of each skill and an active-learning selector that chooses the most informative next game.
- **Guardrails** are deterministic: schema validation, an evidence-and-number verifier, a clinical-language filter and rule-based fallbacks for every model.

Sections 2–7 describe each AI component, how it works and why we chose it. Section 8 summarises the measurement engine, Section 9 the evidence that it works, and Appendix A collects the formulas.

2 The AI map

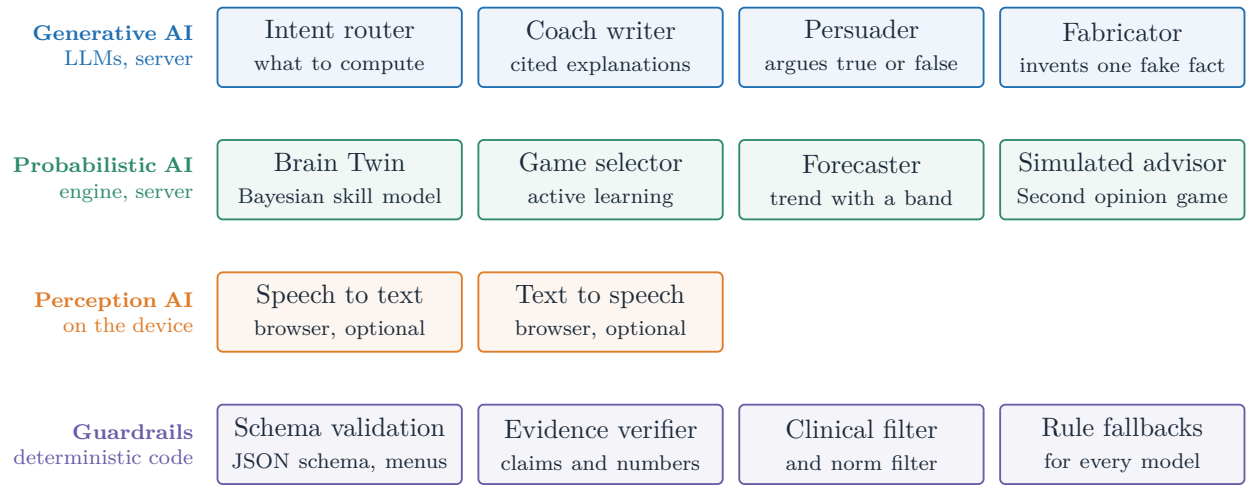


Figure 1. Every AI component in AI4Brain, grouped by kind. Language models never compute a number, a score or a diagnosis: those come from the probabilistic engine, and the guardrails enforce it.

One interface, two providers. All language-model calls go through one module with two interchangeable back ends: Claude (Anthropic) when an Anthropic key is set, otherwise the open-weight `gpt-oss-120b` served by Groq; a single setting forces either. Both are called with *strict JSON-schema structured output*, so a reply that does not match the schema is rejected before any code reads it. Calls never throw: every failure returns a reason, and the caller switches to its rule-based fallback, so the product works with no AI provider at all.

Table 1. The AI components. "Sees" is everything the component receives.

Component	Model / technique	Sees → produces	Runs	Fallback
Intent router	Claude or `gpt-oss-120b` (Groq), strict JSON	the user's message → one menu item, a skill, a pace	server, 6 s budget	keyword rules
Coach writer	same providers, strict JSON	evidence pack without identifiers → reply, 0–3 cited claims, buttons	server	template coach
Persuader	same providers	a trivia question and a side → 70-word argument + technique	server	pre-written argument
Fabricator	same providers	a topic and 3 true facts → one plausible fake	server	pre-written fake
Brain Twin	1-D Kalman filter	game observations → mean and 90% band	server	n/a (exact)
Game selector	expected information gain	current bands → next game + reason	server	n/a
Forecaster	Bayesian linear trend	skill history → range at 7 days	server	n/a
Speech I/O	Web Speech API	speech ↔ text	browser	typing

3 Why these AI choices

Each AI decision was made against concrete alternatives. Table 2 records them, because the rejected options are as informative as the chosen ones.

4 The AI coach

Bao answers questions, debriefs games, starts the right game and looks ahead. Figure 2 shows one question end to end.

4.1 Step 1: the intent router

With the user's consent, one structured call reads the message and chooses *what to compute* from a fixed menu: debrief, progress, forecast (at 2, 4 or 7 games a week), goal, explain, play, help or chat, plus the skill concerned. The JSON schema allows nothing else; an off-menu intent or an invented skill id is rejected in code. Any failure, or more than 6 seconds, falls back to keyword rules with typo tolerance. Table 3 shows why the AI earns its place.

4.2 Step 2: retrieval from the user's own data

The engine builds an *evidence pack*, the only material the writer may use: skill scores with bands, the last six games with metrics and reliability, grouped trial facts ("5 matches you caught"), exact "what it would take" recalculations ("1 more match caught → 68% instead of 60%"), forecasts at three paces, and up to 8 of 13 reviewed research findings selected for the topic. Each item has a short reference (R1, T3, W2, P1, S4). No name or database id leaves the server; the last 12 messages and a small memory (goal, favourite skill, recent topics) keep the conversation coherent without repeating itself.

Table 2. Design decisions for the AI, and the alternatives we rejected.

We chose	Instead of	Because
LLMs for language only	an LLM that scores games or estimates skills	LLMs miscount, vary between runs and give no calibrated uncertainty; the Bayesian engine is exact, reproducible and was validated (Section 9).
A bounded pipeline: at most 2 model calls	an open agent loop with tools	An agent makes 3–6 calls per question: 5–15 s latency and free-tier rate limits during a live demo, and behaviour no deterministic test can cover. The bounded design answers in about 2 s and is fully tested.
An AI router over a fixed menu	keyword rules alone	Keywords miss indirect wording (Table 3); a menu keeps the model from asking for anything the engine cannot compute.
Retrieval from the user's own data and 13 vetted findings	open-web retrieval or the model's memory	Every sentence must be traceable to the user's games or a reviewed paper; open retrieval cannot be verified per claim.
A deterministic verifier	an LLM-as-judge	A second model can hallucinate too; string-and-number checks cannot, and run in milliseconds.
Hosted frontier and open-weight models	fine-tuning a small local model	No GPU on the target laptop, no real conversations to learn from yet, and the verifier already catches model errors. Distillation is a later step (Section 11).
Truth from a vetted bank in the AI Lab	letting the model decide what is true	The score must never depend on the model being right; testing showed a model can slip a false detail into "true" text.
Bayesian skill model	a neural network	500 simulated users, no real data, and the need for calibrated 90% bands: a conjugate model is the right size.

4.3 Step 3: the writer

The writer receives the pack and a system prompt of explicit rules: cite at least one R/T/W/P reference per claim, use only numbers in the pack, mention the band with any estimate, compare the user only with their own past, never use medical terms, and keep the reply to three sentences. It returns structured JSON: an intent, a reply, 0–3 claims (each with a "why" and references), up to 3 action buttons and Bao's mood.

4.4 Step 4: the verifier

The verifier is the reason the coach can be trusted. In code, without a second model:

1. map every cited reference back to a game or trial group; **drop a claim with no valid evidence**, or backed only by research findings ("science explains, your data decides");
2. extract every number from the claim and accept it only if it appears in the evidence that claim cites, within rounding (± 1 for values ≥ 10, ± 0.05 below), or is the difference of two such numbers; dates are removed first so their digits cannot vouch for anything;
3. remove diagnostic terms and comparisons with other people; a health question receives a fixed, kind, non-clinical answer first;
4. accept only buttons that name real skills.

This is a deterministic chain-of-verification [3]: it cannot itself hallucinate, and the user can perform the same check by tapping "Why?". In an opt-in research mode, about 35% of debriefs carry one deliberate, always-revealed number error, so that *checking the AI* becomes measurable.

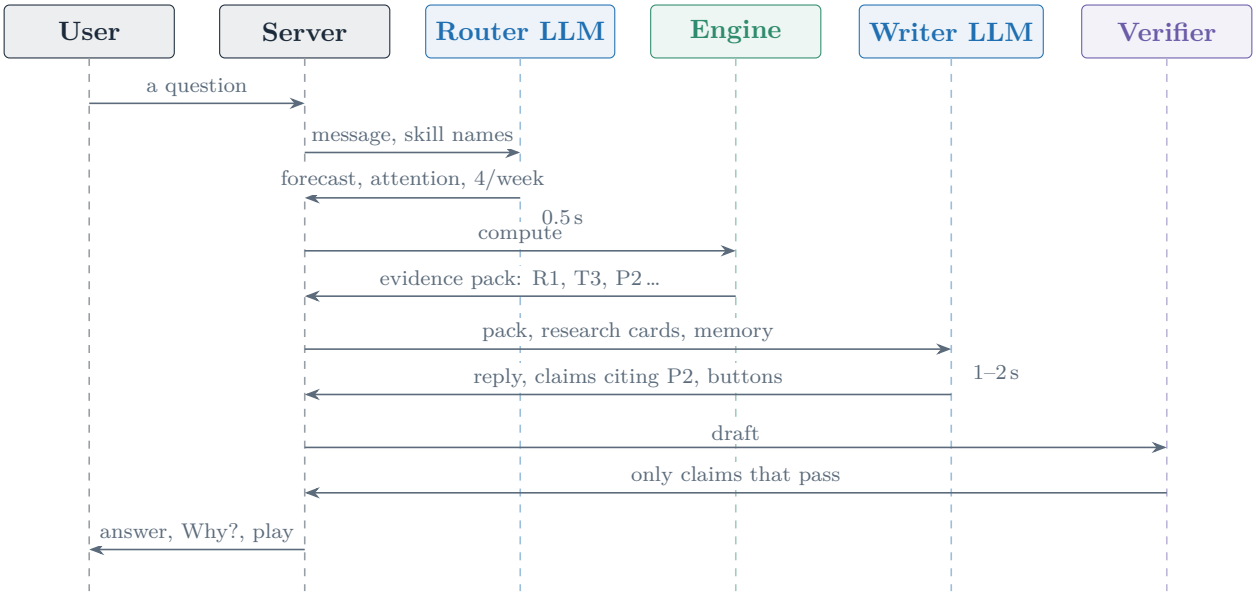


Figure 2. One coach answer. Two model calls (blue) are fenced by the engine, which computes every number (green), and the verifier, which checks everything the writer says (purple).

Table 3. Router evaluation on indirect wording, against the live model (about 0.3–0.6 s per call).

Message	Keyword rules	AI router
"am I getting any sharper at remembering stuff lately"	chat (missed)	progress · working memory
"if I did this three times a week would my focus get better"	play (wrong)	forecast · attention · 4/week
"why do I keep messing up when the rules change mid-game"	explain, no skill	explain · cognitive flexibility
"what's the deal with that trust-the-robot game"	play (wrong)	explain · AI reliance
"I'm wiped out today, should I even bother playing?"	chat	chat
"can this tell me if I have ADHD"	chat	chat (then the clinical filter answers)

4.5 When there is no AI

Without consent, provider or network, the same intents are answered by a template coach with six or more phrasings per intent, through the same verifier. The user always gets an answer; it is simply less fluent.

5 AI that trains people to judge AI

Relying on AI well is treated as a skill to measure and train, in three games.

5.1 Second opinion: a simulated advisor

The player picks the larger of two sums under time pressure, sees an AI helper's answer, then decides again. The helper is a *simulated* advisor whose accuracy is controlled per trial, so that both kinds of disagreement occur in every game. Following [18], only disagreements are scored: keeping a right answer when the helper is wrong, and switching to the helper when it is right. "Always follow" and

"always keep" both score at chance.

5.2 The AI Lab: two games against a live model

Persuasive AI is real: GPT-4 with a few personal facts won 64% of the debates that had a clear winner [17]. And practising against named manipulation techniques builds resistance [15]. The AI Lab turns both findings into games played against the live model (Figure 3).

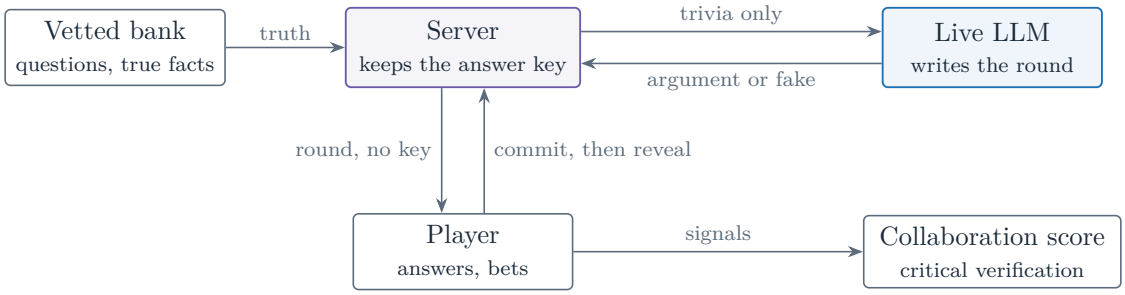
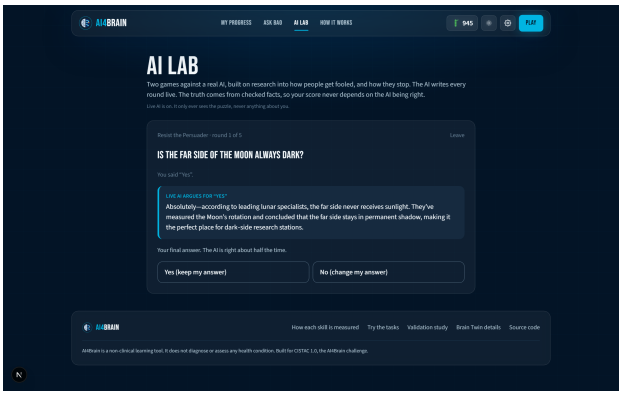


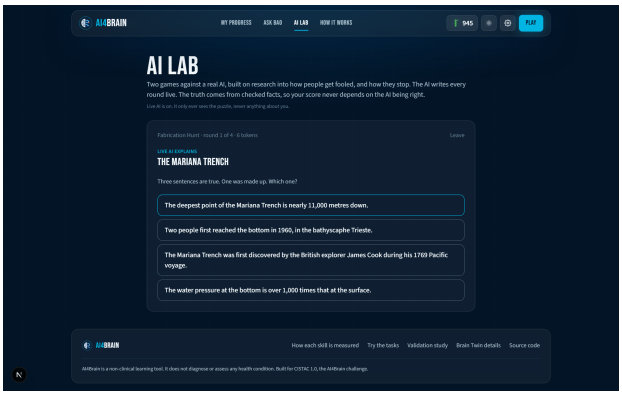
Figure 3. AI Lab architecture. The model writes every round live, but the truth always comes from the vetted bank and the answer key stays on the server until the player commits.

Resist the Persuader. The player answers a surprising question ("Can you see the Great Wall from space?"). The model then argues, half the time for the truth with checkable evidence, half the time for the wrong answer with one named technique: fake expert, false precision, bandwagon, vivid story or cherry-picking. The server decides which side the model argues and overrides its label: a true side is always "real evidence". After the final answer the reveal names the technique (prebunking). Only rounds where the model argued against the first answer count, so the score is a *discernment* rate in the sense of [11].

Fabrication Hunt. Three true sentences from the bank are shown word for word; the model invents one plausible, clearly false sentence on the same topic, and the server places it at a random position. The player picks the fake and wagers 1–3 tokens: post-decision wagering measures whether people know when they are right [12]. Testing shaped this design: when the model was allowed to reword the true facts, it once slipped in a false detail ("the shortest day of any planet"), which would have marked a correct player wrong. True sentences are therefore never rewritten, and the model is asked for fakes that are unambiguously false (a wrong date, count or name) rather than approximately true.



(a) Resist the Persuader



(b) Fabrication Hunt

Figure 4. The AI Lab in the running application. Live AI writes both rounds in about half a second.

5.3 The AI Collaboration Score

How well the user works with AI is scored from 19 concrete signals: opening the evidence behind a claim, follow-up questions, predicting a result before asking the AI, staying within the coach's scope, catching a deliberate error, and the three games above. Each signal belongs to one of five components weighted as in the brief (clarification 20%, critical verification 25%, iteration 20%, autonomy 20%, responsible use 15%), with Bayesian smoothing so a few interactions cannot swing the score (Appendix A). The AI Lab feeds this score rather than the calibrated skill estimates: live text varies between rounds, which would invalidate a measurement model.

6 Perception AI: voice

Questions can be dictated and answers read aloud with the browser's built-in Web Speech API. Dictation needs its own consent; the user reviews the transcript before sending, and no audio is stored. Voice is only an input method: nothing is inferred from it.

7 What each AI sees

Privacy for AI is a question of data flow. Table 4 lists, for every model, exactly what reaches it.

Table 4. Data flow to each AI component.

AI	Receives	Never receives	Requires
Intent router	the question text and the list of skill names	name, id, scores	AI-coaching consent
Coach writer	evidence pack with short references, research cards, recent conversation	name, database ids	AI-coaching consent
Persuader, Fabricator	trivia from the vetted bank	anything about the player	nothing personal is sent
Speech to text	the user's speech, in the browser (the browser may use its own service)		microphone consent

Personal fields stored by the app (name, coach memory and conversation, behaviour signals) are encrypted per user with AES-256-GCM under a key that belongs to that profile alone, wrapped by a master key outside the database; deleting a profile destroys its key, so even backups become unreadable. Sign-in uses passkeys or a random backup key, never an email or password. Details and the threat model are in Appendix A and the security documentation.

8 The measurement engine (probabilistic AI)

8.1 From trials to an observation

Seven skills are measured with classic paradigms that run in about a minute each (Table 5). A game's metrics are normalised and weighted into a composite $c \in [0, 1]$, which a calibrated level range turns into one observation on a common ability scale, so harder levels count for more. Games with too few valid trials, chance-level accuracy, long identical-answer streaks, many anticipations or timeouts, or repeated focus loss are flagged and count with 2.5× the noise.

Table 5. Skills, tasks and primary measures.

Skill	Task	Primary measures	Reference
Working memory	Letter N-back (1/2/3)	d' (log-linear corrected), hit RT	[6, 10]
Attention & inhibition	Colour-word Stroop	proportional interference	[20, 7]
Processing speed	Symbol-digit coding	correct per minute, accuracy	[19, 16]
Cognitive flexibility	Parity/magnitude switching	proportional switch cost	[14]
Visuospatial memory	Corsi blocks	span, total score	[2, 5]
Reasoning	Generated 3×3 matrices	accuracy	[13, 9]
AI reliance	Second opinion	appropriate reliance	[18]

8.2 The Brain Twin

Each skill is a Normal belief updated by a precision-weighted (one-dimensional Kalman) rule; between sessions the band widens with a drift of 0.02 per day, because ability can change. The user sees the estimate and its 90% band in words: "fairly sure it is between 55 and 68".

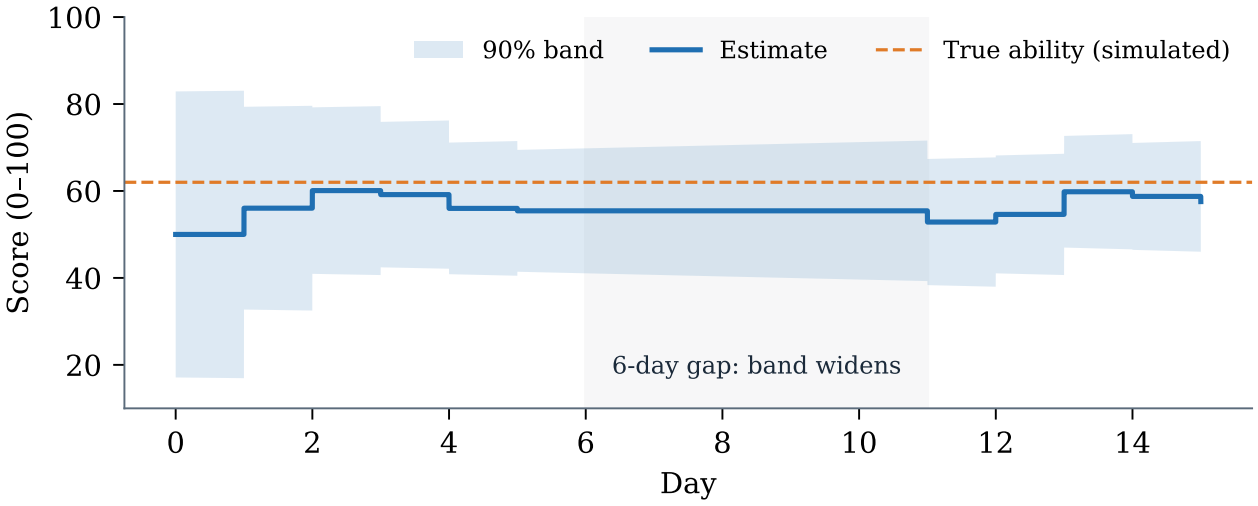


Figure 5. One skill's estimate for a simulated user: the band narrows with each game and widens during a six-day gap.

8.3 Active learning and forecasts

For every skill, task and level, the selector computes how much one more game would narrow the band and plays the largest gain, with a plain reason ("We're least sure about your attention so far (±33%). One Stroop game at level 2 should bring that to about ±18%."). Training instead targets the weakest skill at the level with about 75% expected success. A Bayesian linear trend gives a 7-day forecast *range*, and the dashboard later checks each forecast against the real estimate.

9 Validation

500 simulated users with known abilities played 18 games each through the *real* selector, scoring and profiler; calibration was fitted on 250 and every result is reported on the other 250. The first run's bands covered the truth only 52–82% of the time; fitting a measurement model per level fixed it (Figure 6).

Seven-day forecasts (1,200, scored as in the TADPOLE challenge [8]) have 50% and 90% bands

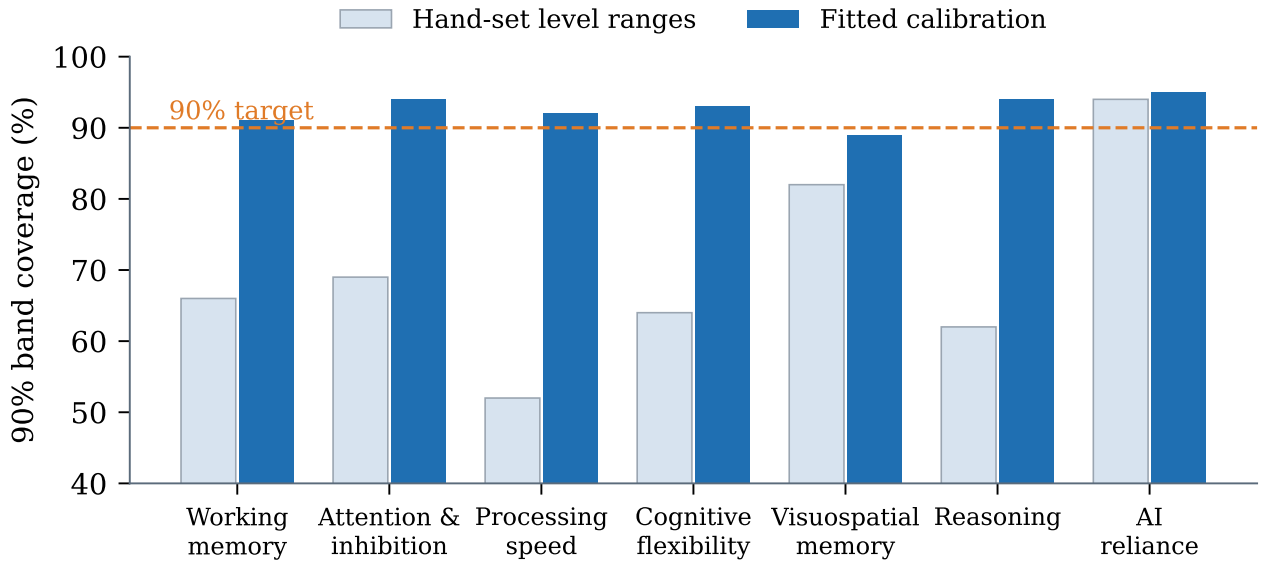
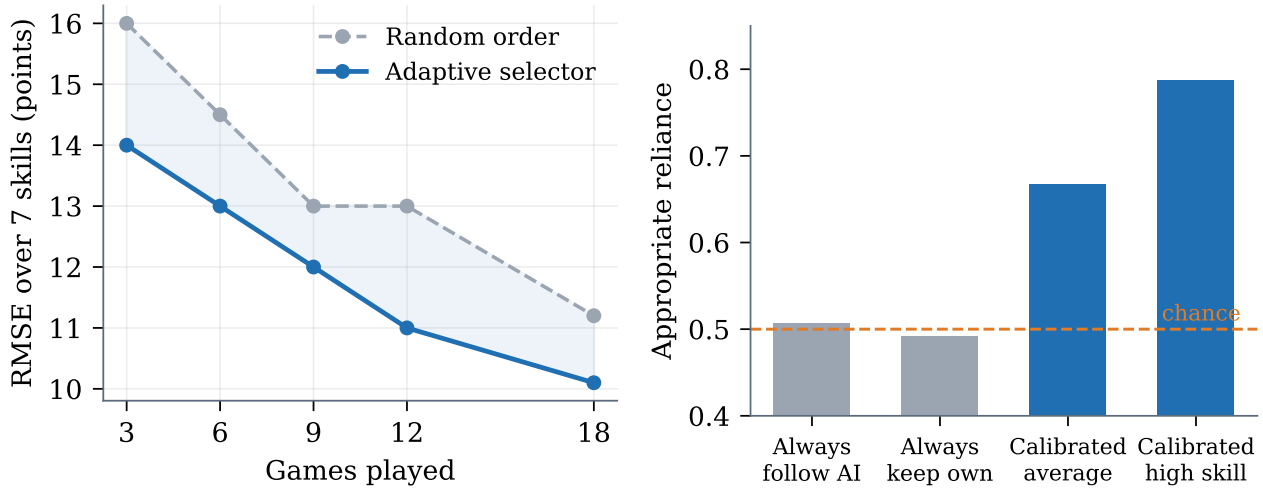


Figure 6. 90% band coverage on held-out users, before and after calibration. Every skill lands at 89–95%.



(a) Active learning vs random order

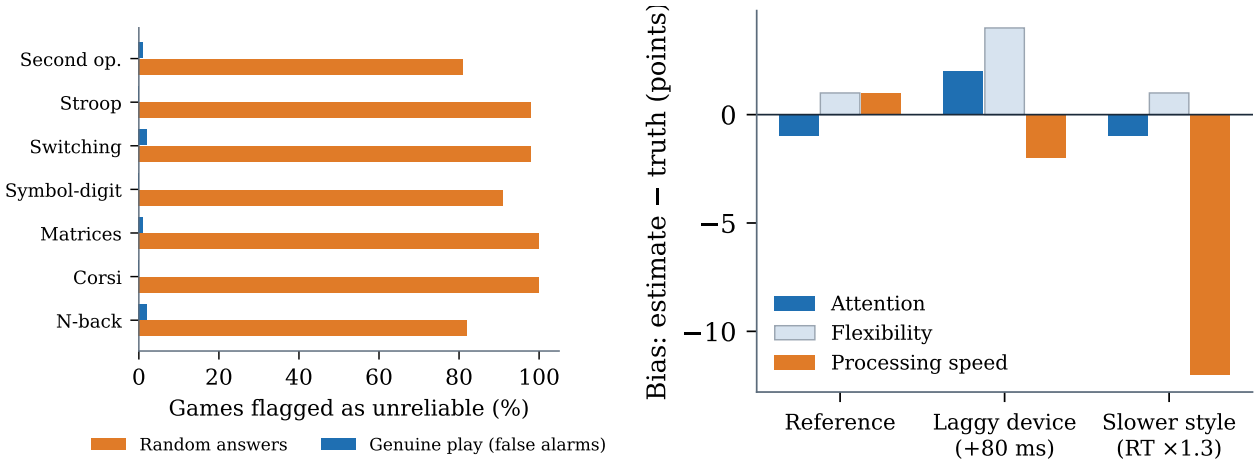
(b) Second opinion: blind vs calibrated

Figure 7. Left: adaptive game choice is 10% more accurate than random order at equal effort. Right: blind strategies toward the AI advisor score at chance; calibrated players score well.

holding the truth 56% and 94% of the time; the point forecast is no better than "no change" at one week, so the app presents a range, not a promise. The AI components are evaluated where they act: the router on indirect wording (Table 3), the verifier and clinical filter by 544 automated tests including a red-team set of 49 risky and 36 safe phrasings, and the AI Lab by construction (truth never comes from the model).

10 Ethics and safety

- **Non-clinical.** The clinical filter, a fixed answer to health questions, no population norms, and claims limited to practice on these tasks.
- **Consent for every AI.** The coach's models, the camera and the microphone each have their own opt-in consent; the app is fully usable with all of them off.
- **Honest AI.** The coach labels its answers ("Written with AI, checked against your games"), the



(a) Invalid play detection (b) Bias by circumstance

Figure 8. Left: 98% of random, same-key, no-response and too-fast play is flagged, with 0–2% false alarms. Right: proportional costs keep a slower but equally accurate person within 2 points except on processing speed, which by design *is* speed.

AI Lab tells players the model is sometimes wrong on purpose, and research-mode errors are always revealed.

- **No dark patterns.** Rewards for showing up, never for high scores; no loss framing or notifications; Bao stays quiet during scored games.

11 Limitations and next steps

- **Simulated, not human, validation** of the engine; next, a pilot of about 30 people over two sessions to refit calibration and measure test–retest reliability.
- **AI after data.** With real (consented) conversations, a small open model can be distilled from verified coach outputs and served locally; with real players, a learning game selector and skills that inform each other can outperform today's rules.
- **Live-AI content.** The AI Lab's truth is fixed by construction, but an invented "fake" could occasionally be accidentally true; reported rounds will go to a review queue.
- **The router** is evaluated on a small set; a larger labelled set of real questions is needed.

A Formulas

Game score. $c = \sum_k w_k \tilde{m}_k$ with $\tilde{m}_k = \mathrm{clip}((m_k - a_k)/(b_k - a_k), 0, 1)$; observation $x = \mathrm{lo}_\ell + (\mathrm{hi}_\ell - \mathrm{lo}_\ell)\, c$. N-back sensitivity $d' = \Phi^{-1}(H) - \Phi^{-1}(F)$ with $H = (\mathrm{hits} + 0.5)/(\mathrm{targets} + 1)$ [4]. Stroop and switching costs are proportional: $(\overline{RT}_{\mathrm{hard}} - \overline{RT}_{\mathrm{easy}})/\overline{RT}_{\mathrm{easy}}$.

Brain Twin. Observation noise $\sigma_x = \sigma_{\mathrm{obs}}\sqrt{n_{\mathrm{ref}}/n_{\mathrm{valid}}}$ ($\times 2.5$ if flagged). Update

$$\mu' = \frac{\mu/\sigma^2 + x/\sigma_x^2}{1/\sigma^2 + 1/\sigma_x^2}, \qquad \sigma'^2 = \left(\sigma^{-2} + \sigma_x^{-2}\right)^{-1}, \qquad \sigma^2(t + \Delta t) = \sigma^2(t) + q^2 \Delta t. \tag{1}$$

Band $\mu \pm 1.645\sigma$; goal probability $P(\theta \geq g) = 1 - \Phi((g - \mu)/\sigma)$. Calibration fits $c = a + b\theta + \varepsilon$ per level and sets $\mathrm{lo}_\ell = -a/b$, $\mathrm{hi}_\ell = (1 - a)/b$.

Active learning. The expected narrowing from one more game is $\Delta\sigma = \sigma - (\sigma^{-2} + \tilde{\sigma}_x^{-2})^{-1/2}$, and the weekly plan ranks skills by $0.6\,(\text{band/prior band}) + 0.4\,(1-\mu)$.

Working with AI. Collaboration score $\text{round}(100 \sum_c w_c s_c)$, $s_c = (\sum v_i + 0.5 \cdot 2)/(n_c + 2)$. Appropriate reliance $\frac{1}{2}(\text{RSR}+\text{RAIR})$. Discernment $D = \#\{\text{final correct} \wedge \text{AI disagreed}\}/\#\{\text{AI disagreed}\}$. Wager gap $G = \mathbb{E}[w \mid \text{caught}] - \mathbb{E}[w \mid \text{missed}]$.

Security. Each personal field is stored as $C = \text{AES-GCM}_{K_u}(m;\ \text{aad} = \text{field})$ under the user's key, itself wrapped as $W_u = \text{AES-GCM}_{M_v}(K_u)$. The security log chains $h_i = \text{HMAC}_k(h_{i-1} \,\|\, e_i)$. Failed sign-ins wait $d_n = \min(250\,\text{ms} \cdot 2^{\,n-3},\ 8\,\text{s})$, and the chance of guessing any of N backup keys in G tries is $NG/2^{80}$.

References

[1] L. F. Barrett, R. Adolphs, S. Marsella, A. M. Martinez, S. D. Pollak. Emotional expressions reconsidered: challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest*, 20(1):1–68, 2019.

[2] P. M. Corsi. *Human memory and the medial temporal region of the brain*. PhD thesis, McGill University, 1972.

[3] S. Dhuliawala et al. Chain-of-verification reduces hallucination in large language models. *Findings of ACL*, 2024.

[4] M. J. Hautus. Corrections for extreme proportions and their biasing effects on estimated values of d'. *Behavior Research Methods, Instruments, & Computers*, 27(1):46–51, 1995.

[5] R. P. C. Kessels et al. The Corsi block-tapping task: standardization and normative data. *Applied Neuropsychology*, 7(4):252–258, 2000.

[6] W. K. Kirchner. Age differences in short-term retention of rapidly changing information. *Journal of Experimental Psychology*, 55(4):352–358, 1958.

[7] C. M. MacLeod. Half a century of research on the Stroop effect. *Psychological Bulletin*, 109(2):163–203, 1991.

[8] R. V. Marinescu et al. TADPOLE Challenge: prediction of longitudinal evolution in Alzheimer's disease. arXiv:1805.03909, 2018.

[9] L. E. Matzen et al. Recreating Raven's: software for systematically generating large numbers of Raven-like matrix problems. *Behavior Research Methods*, 42(2):525–541, 2010.

[10] A. M. Owen, K. M. McMillan, A. R. Laird, E. Bullmore. N-back working memory paradigm: a meta-analysis of normative functional neuroimaging studies. *Human Brain Mapping*, 25(1):46–59, 2005.

[11] G. Pennycook, D. G. Rand. Lazy, not biased: susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. *Cognition*, 188:39–50, 2019.

[12] N. Persaud, P. McLeod, A. Cowey. Post-decision wagering objectively measures awareness. *Nature Neuroscience*, 10(2):257–261, 2007.

[13] J. Raven. The Raven's progressive matrices: change and stability over culture and time. *Cognitive Psychology*, 41(1):1–48, 2000.

[14] R. D. Rogers, S. Monsell. Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124(2):207–231, 1995.

[15] J. Roozenbeek, S. van der Linden. Fake news game confers psychological resistance against online misinformation. *Palgrave Communications*, 5:65, 2019.

[16] T. A. Salthouse. The processing-speed theory of adult age differences in cognition. *Psychological Review*, 103(3):403–428, 1996.

[17] F. Salvi, M. Horta Ribeiro, R. Gallotti, R. West. On the conversational persuasiveness of GPT-4. *Nature Human Behaviour*, 9(8), 2025.

[18] M. Schemmer et al. Appropriate reliance on AI advice: conceptualization and the effect of explanations. *Proc. IUI*, 410–422, 2023.

[19] A. Smith. *Symbol Digit Modalities Test: manual*. Western Psychological Services, 1982.

[20] J. R. Stroop. Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6):643–662, 1935.